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Bihar Engineering University, Patna

B.Tech. 1st Semester Examination, 2023

Subject: Mathematics-1 (Calculus and Linear Algebra)

Code: 105102

Time: 03 Hours

Full Marks: 70

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. 1 is compulsory.*
(v) *Symbols used (if any) have their usual meanings.*

Q.1 Choose the correct answer of the following (Any seven question only): **[14]**

(a) The value of $\int_0^1 x^3(1-x^2)^{5/2}dx$ is

(i) $\pi/2$
(ii) 2/63
(iii) 1
(iv) None of the above

(b) The area $\iint_R \sin(x+y)\,dxdy$ over R $\{(x, y); 0\le x\le\pi/2, 0\le x\le\pi/2\}$

(i) 0
(ii) 1
(iii) 2
(iv) -1

(c) The matrix $\begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix}$ is

(i) Invertible but not orthogonal
(ii) Invertible and orthogonal
(iii) Neither invertible nor orthogonal
(iv) None of the above

(d) The sequence $\{a_n\}$, where $a_n=\text{sqrt}[n]n$ is

(i) Is not convergent.
(ii) Convergent and $\lim_{n\to\infty} a_n = 0$
(iii) Convergent and $\lim_{n\to\infty} a_n = 1$
(iv) None of the above

(e) Let $f(x) = \cos(x^2)$, $X \text{in} R$ then

(i) f is uniformly continuous
(ii) f is continuous, but not uniformly continuous
(iii) f continuous but unbounded
(iv) f is Lipschitz continuous

(f) $\lim_{(x,y)\to(1,2)} [x^2+ y] / [x+y^2]$

(i) Does not exists
(ii) Exists and the value is 2
(iii) Exists and the value is 3
(iv) None of the above

(g) $A = 2i + \alpha j + k$, $B = i + 3j - 8k$. Then A and B are orthogonal if

(i) $\alpha = -2$
(ii) $\alpha = 2$
(iii) $\alpha = -1$
(iv) $\alpha = 1$

(h) Let $x + 2y + z = 0$, $3x + 4y + z = 0$, $x - z = 0$ be a system of equations. Then

(i) It is inconsistent
(ii) It has only the trivial solution $x = 0$, $y = 0$, $z = 0$
(iii) Determinant of the matrix of coefficient is zero
(iv) None of these

(i) The value at which the Rolle's theorem is applicable for $f(x) = \cos x/2$ in $|\pi, 3\pi|$ is

(i) $5\pi/2$
(ii) $3\pi/2$
(iii) 2π
(iv) none of the above

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(j) The rank of the matrix $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{vmatrix}$ is

(i) 3
(ii) 2
(iii) 1
(iv) none of these

Q.2 Solve both questions : **[14]**

(a) Show that: $\int_0^{\pi/2} \log(\tan x + \cot x)\, dx = \pi \log 2$

(b) Find the following limit: $\lim_{n \to \infty}[\, t \ln(1 + 3/t)]$

Q.3 Solve both questions : **[14]**

(a) Test the convergence of $x^2 + 2^2/3.4x^4 + [2^2 4^2] / [3.4,5.6,7.8]x^8 + cdots$

(b) Examine the convergence of the series of which the general term is $\sum_{n=1}^{\infty}[1] / [n(n+1)]$

Q.4 Solve both questions : **[14]**

(a) Prove that $\log(1+x) = [x] / [1+\theta x]$, where $0 < \theta < 1$. Hence deduce, $[x] / [1+x] < \log(1+x) < x$

(b) It is given the Rolle's theorem holds the function $f(x) = x^3 + bx^2 + cx$, $1 \le x \le 2$ at the point $x = 4/3$. Find the value of b and c.

Q.5 Solve both questions : **[14]**

(a) Evaluate $\int_0^{\pi/2}[1] / [a\cos^2 x + b\sin^2 x]\, dx$

(b) Discuss the convergence of the sequence whose n-th term is $a_n = [(-1)^n] / [n] + 1$.

Q.6 Solve both questions : **[14]**

(a) Show that the equation of the evolute of the curve $x^{2/3} + y^{2/3} = a^{2/3}$ is $(x+y)^{2/3} + (x-y)^{2/3} = 2a^{2/3}$.

(b) Find the areas of the regions that lies inside the circle $r = a\cos\theta$ and outside the cardioid $r = a(1 - \cos\theta)$.

Q.7 Solve both questions : **[14]**

(a) Find the solution of the following system of equations:
$x_1 + 2x_2 + 2x_3 = 2$
$x_1 + 8x_3 + 5x_4 = -6$
$x_1 + x_2 + 5x_3 + 5x_4 = 3$
Also find the basis and dimension of the solution space.

(b) Verify Cayley-Hamilton theorem for the matrix $A = \begin{vmatrix} 2 & 4 \\ 2 & 5 \end{vmatrix}$ and find its inverse. Also express $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$ as a linear polynomial in A.

Q.8 Solve both questions : **[14]**

(a) Expand $\sin x$ in Taylor's series about $x = \pi/2$.

(b) By using Beta function evaluate $\int_0^1 x^4(1 - \sqrt{(x)})^5\, dx$.

Q.9 Solve both questions : **[14]**

(a) Develop the Fourier series on $-\pi < x < \pi$ to represent the function defined as:
$f(x) = 0$, & $-\pi < x < 0$
π, & $0 < x < \pi$

(b) (i) Evaluate div F and Curl F, where $F = grad(x^2 + y^2 + z^2)\, e^{-\sqrt{(x^2+y^2+z^2)}}$.
(ii) Let $F = (-4x - 3y + az)\, i + (bx + 3y + 5z)\, j + (4x + cy + 3z)\, k$. Find the values of a, b and c such that F is irrotational.

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